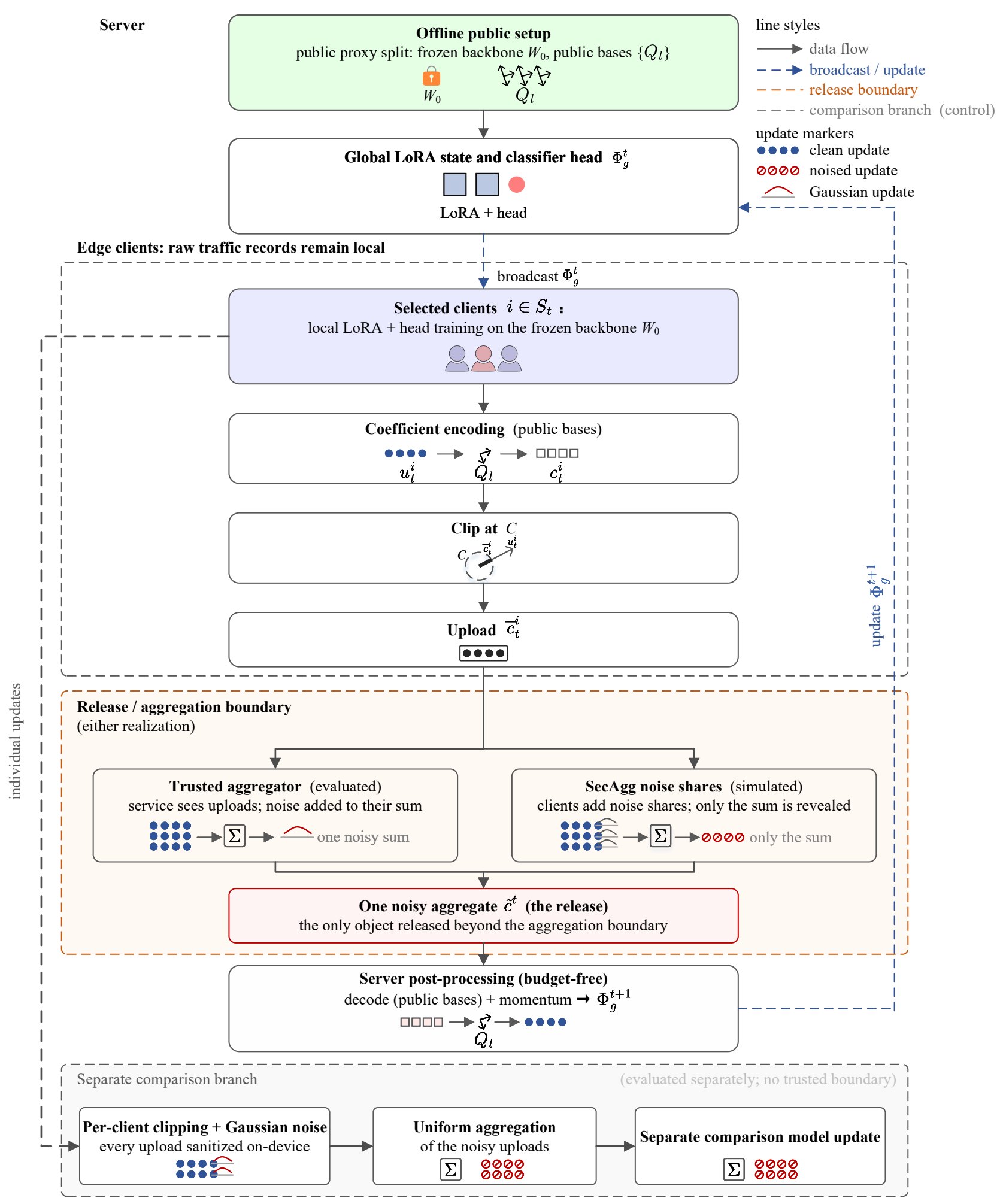

Server
Offline public setup
public proxy split: frozen backbone W_0, public bases $\{Q_l\}$
W_0
Q_l
line styles
data flow
broadcast / update
release boundary
comparison branch (control)
update markers
clean update
noised update
Gaussian update
Global LoRA state and classifier head Φ_g^t
LoRA + head
Edge clients: raw traffic records remain local
broadcast Φ_g^t
Selected clients $i \in S_t$:
local LoRA + head training on the frozen backbone W_0
Coefficient encoding (public bases)
u_t^i
Q_l
c_t^i
Clip at C
C
$\tilde{c}_t^i$
u_t^i
Upload $\tilde{c}_t^i$
update Φ_g^{t+1}
individual updates
Release / aggregation boundary
(either realization)
Trusted aggregator (evaluated)
service sees uploads; noise added to their sum
one noisy sum
SecAgg noise shares (simulated)
clients add noise shares; only the sum is revealed
only the sum
One noisy aggregate $\tilde{c}^t$ (the release)
the only object released beyond the aggregation boundary
Server post-processing (budget-free)
decode (public bases) + momentum → Φ_g^{t+1}
Q_l
Separate comparison branch
(evaluated separately; no trusted boundary)
Per-client clipping + Gaussian noise
every upload sanitized on-device
Uniform aggregation
of the noisy uploads
Separate comparison model update